# CroMaSt: A workflow for domain family curation through cross-mapping of structural instances between protein domain databases

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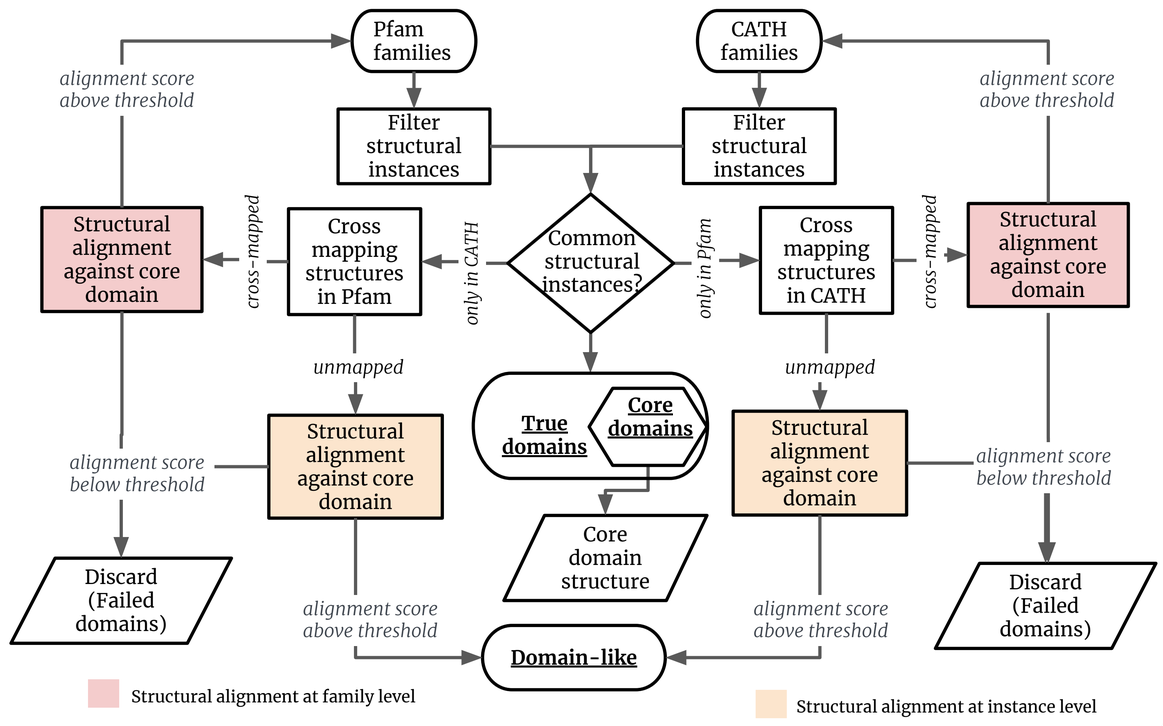
## Overview

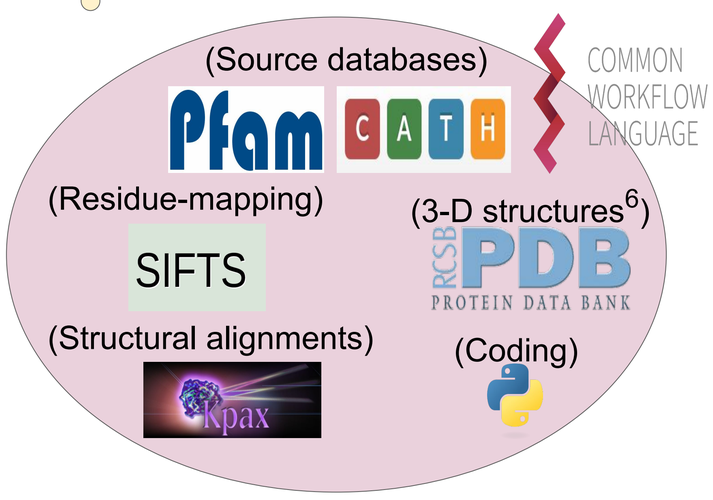
Protein domains are building blocks of proteins, essential for understanding structure-function relationships. However the diversity of domain databases makes it difficult to capture a clear definition for a given type of domain and to enumerate its true instances.

CroMaSt (**Cro**ss-**Ma**pper for **St**ructural domains) is an automated iterative workflow to clarify domain definition by cross-mapping of domain structural instances between domain databases. It classifies all structural instances of a given domain type into 3 different categories (core, true and domain-like). CroMaSt is developed in Common Workflow Language (CWL)[1] and takes advantage of 2 well-known and widely used domain databases, Pfam[2] (sequence-based) and CATH[3] (structure-based).

| Category | Starting family member | Cross-mapped | Structurally well aligned |
|---|---|---|---|
| Core | ✓ | ✓ | ✓ |
| True | - | ✓ | ✓ |
| Domain-like | - | - | ✓ |




## Implementation

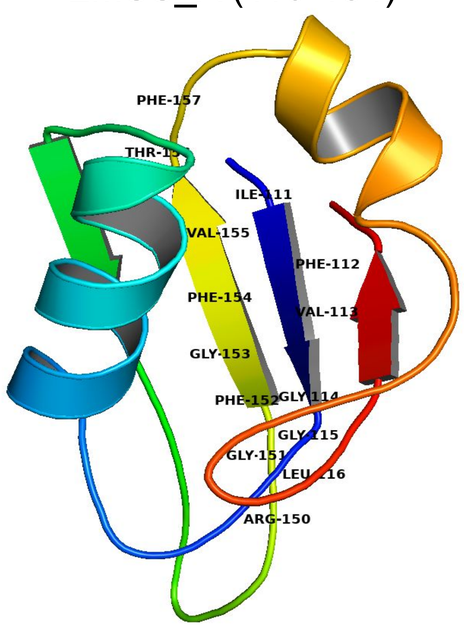



**Main difficulties:**

- Residue-mapping (SIFTS[4])
- Cross-mapping (Source db)
- Structural alignments (Kpax[5])
- Computing average structures (3-D coordinates)

## Structural instance

2MSS_A (110-184)




Pfam: PF00076 (RRM_1)
CATH: 3.30.70.330 (RRM)

## Results

| Steps | Iteration 1 Pfam | Iteration 1 CATH | Iteration 2 Pfam | Iteration 2 CATH |
|---|---|---|---|---|
| Starting Families | 1 | 1 | 14 | 0 |
| StIs filtered on domain length | 1147 | 1527 | 96 | 80* |
| Obsolete and inconsistent entries | 3 | 323 | 0 | 0 |
| **=>Residue-mapped StIs** | 1144 | 1204 | 96 | – |
| Common StIs (**Core** & **True**) | 886 | 886 | 80 | 80 |
| Remaining StIs (not common) | 258 | 318 | 16 | 0 |
| Cross-mapped StIs | 0 | 244 | 0 | 0 |
| Properly aligned at family level | – | 80 | – | – |
| Not properly aligned at family level | – | 164 | – | – |
| Not cross-mapped StIs (unmapped) | 258 | 74 | 16 | 0 |
| Properly aligned at instance level (**Domain-like**) | 255 | 74 | 15 | – |
| Not properly aligned at instance level | 3 | 0 | 1 | – |
| Failed structural instances | 3 | 164 | 1 | 0 |
| New families found | 14 | 0 | 0 | 0 |

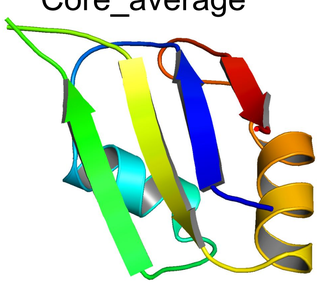
*StI: Structural instance*

*These StI entries are cross-mapped and properly aligned at the family level from previous iteration

## Results Visualization

### Core domain

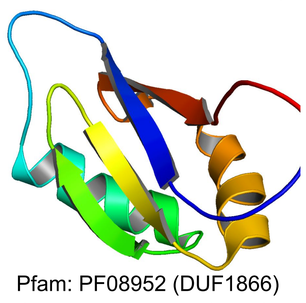
Core_average



Pfam: PF00076 (RRM_1)
CATH: 3.30.70.330 (RRM)

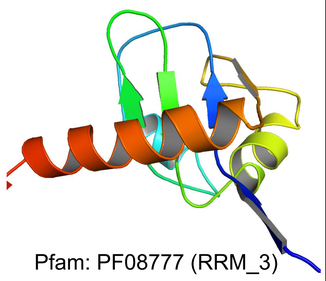
### True-domains

2DNR_A (886-976)



Pfam: PF08952 (DUF1866)
CATH: 3.30.70.330 (RRM)

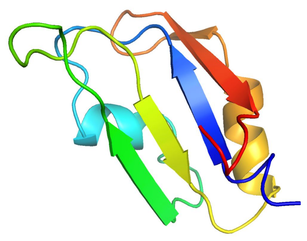
1OWX_A (222-334)



Pfam: PF08777 (RRM_3)
CATH: 3.30.70.330 (RRM)

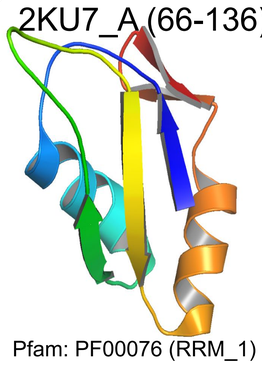
### Domain-like

2DNN_A (230-311)



CATH: 3.30.70.330 (RRM)

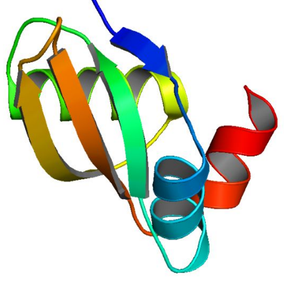
2KU7_A (66-136)



Pfam: PF00076 (RRM_1)

### Failed domains

3CCE_S (1-81)



CATH: 3.30.70.330 (RRM)

All 3-D structure visualizations were generated using PyMol[7]

## Conclusion & Perspectives

- Generic, can be used for other domain types
- Protein design: Provides the prototype for given domain type
- Database curation: Explores "allowed" domain diversity and detects inconsistencies

Complete Workflow Visualize

## References

1. Michael R. Crusoe, doi:10.1145/3486897; 2022
2. Mistry, Jaina et al. doi:10.1093/nar/gkaa913; 2021
3. Sillitoe, Ian et al. doi:10.1093/nar/gkaa1079; 2021
4. Dana, Jose M et al. doi:10.1093/nar/gky1114; 2019
5. Ritchie, David W. doi:10.1093/bioinformatics/btw300; 2016
6. Burley, Stephen K et al. doi:10.1093/nar/gkaa1038; 2021
7. PyMOL, Version 2.5.2, Schrödinger, LLC

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